

HW06

Sam Ly

March 22, 2026

Exercise 1

Collaborators: Sam Ly

Tools and resources used: On Exercise 1, I just copied from the Assignment page.

Exercise 3

- Points attempted: 5
- Self-assessed score: 5/5.
- Time spent: 20 minutes.
- Difficulty: 3/10.
- Questions/Feedback: None.

Give examples of each of the following:

1. one group of order 1;
The trivial group $\{e\}$.
2. one group of order p for each prime p ;
The cyclic groups C_p .
3. two groups of order 4 (both abelian);
We have cyclic group C_4 and the group of even integers $2\mathbb{Z}/8$.
4. two groups of order 6 (one is abelian);
We have the groups C_6 and D_3 .
5. five groups of order 8 (three are abelian, and one group we saw on Exam 1);
We have the three abelian groups C_8 , $C_4 \times C_2$, $C_2 \times C_2 \times C_2$, and the two non-abelian groups D_4 and the Quaternion group.
6. two groups of order 9 (both are abelian);
We have the groups C_9 and $C_3 \times C_3$.
7. two groups of order 10 (one is abelian).
We have the the group C_{10} and D_5 .

Choice Exercise 4

- Points attempted: 5
- Self-assessed score: 5/5.
- Time spent: 5 minutes.
- Difficulty: 1/10.
- Questions/Feedback: None.

1. (Do this part after you do the second part.) Write at least three sentences answering the following.

(a) What are a few ideas from HW 4 Exercise 2 that make a lot of sense to you?

Part 1.b made a lot of sense to me, as I was able to relate it back to a computer science concept I have learned previously.

(b) What ideas did you find most difficult?

The most difficult problem for me was 2.d because I can ‘physically’ see that the elements do not commute, but it is difficult to form an intuition for why, and what this implies.

(c) Can you think of any ways to generalize the definitions or propositions that appeared in that problem?

We can likely generalize the rule of all dihedral groups D_n can be written in the form $r^i f^j$ to some other types of groups that have more than two ‘kinds’ of elements. For example, $a^i b^j c^k$, for some group generated by $\{a, b, c\}$ with some relations $a^n = e, b^3 = e, c^2 = e$. I am struggling to really articulate this, but it ‘feels’ like there is something there.

2. Copy/paste your answer from HW 4, re-read it, and make at least three changes that improve the exposition or correctness of the proofs.

I changed two paragraphs of a.i, and one paragraph of a.iii.

(a) We defined the dihedral group in terms of its group presentation:

$$D_n = \langle r, f \mid r^n = f^2 = (rf)^2 = e \rangle.$$

i. Prove that $r^k f = f r^{n-k}$ for all $0 < k < n$.

Proof. First, we notice that $r f r f = e$. By right multiplying by f we get $r f r = f$. By left multiplying by r^{-1} , we get $f r = r^{-1} f$. Then, we left and right multiply by f on both sides to get $r f = f r^{-1}$. *I fixed a type here.*

I added more reasoning to improve logical flow. We can continue this process for $r^n f r^n f = e$ to show that $r^1 f = f r^{n-1}$. Now, to prove that $r^k f = f r^{n-k}$ for all $0 < k < n$, we must use induction. We have already proved the base case $k = 1$. As our inductive hypothesis, we assume that the proposition is true for some value k . Thus, $r^k f = f r^{n-k}$. Now, for by left multiplying by r , we get

$$r^{k+1} f = r f r^{n-k}.$$

Now, we apply $r f = f r^{n-1}$ to the RHS, and get

$$r^{k+1} f = f r^{-1} r^{n-k} = f r^{n-(k+1)}.$$

Therefore, $r^k f = f r^{n-k}$ for all $0 < k < n$. □

ii. Prove that every element can be written in the form $r^k f^\ell$ for $0 \leq k < n$ and $\ell \in \{0, 1\}$

Proof. We approach this problem from a formal languages angle. We begin by constructing all possible ‘words’ of the language formed by the alphabet $\{r, s\}$. One example of such a word is $rrrfrfffrfrrrrrrrf$. By definition, this language must encompass all of D_n for any n .

Then, we can form some basic substitution rules, in other words, a ‘grammar’ that allows us to rewrite the word into a nicer form. By definition of the dihedral group, we have the rules:

$$\underbrace{rr \dots r}_{m \text{ terms}} = r^m = r^{m-n} \quad (1)$$

$$ff = e. \quad (2)$$

Thus, we can rewrite our word into the form $r^{k_0} f^{l_0} r^{k_1} f^{l_1} \dots$, where $0 \leq k_i < n$ and $l_i \in \{0, 1\}$. Now, from here, we apply the proposition proved above to combine and move all l_i to the right of the expression.

For example, if $l_0 = 0$, then the expression becomes $r^{k_0} r^{k_1} f^{l_1} \dots = r^{k_0+k_1} f^{l_1} \dots$. If $l_0 = 1$, then the expression becomes

$$r^{k_0} (f r^{k_1}) f^{l_1} \dots = r^{k_0} (r^{n-k_1} f) f^{l_1} \dots$$

which we can simplify by combining ff^{l_1} . This process can be repeated until we reach the form $r^k f^\ell$ for $0 \leq k < n$ and $\ell \in \{0, 1\}$. $\square$

iii. Prove that the order (number of elements) of the group is $D_n = 2n$.

Proof. Now that we have established that all elements of D_n can be written as the form $r^k f^\ell$ for $0 \leq k < n$ and $\ell \in \{0, 1\}$, this question can be reframed as asking “How many unique ways can we write $r^k f^\ell$?”

I made this statement more detailed. Thus, for k , we have n possible choices, and for ℓ , we have 2 possible choices. Therefore, there are $n * 2 = 2n$ total choices possible. Thus, $|D_n| = 2n$. $\square$

(b) We say that a group $(G, *)$ is an abelian group if and only if $a * b = b * a$ for all $a, b \in G$.

i. Prove that if G is generated by g_1 and g_2 then G is abelian if and only if $g_1 * g_2 = g_2 * g_1$.

Proof. First, we prove that if $a * b = b * a$ for all $a, b \in G$, then $g_1 * g_2 = g_2 * g_1$. By setting $a = g_1$ and $b = g_2$, then $g_1 * g_2 = g_2 * g_1$.

The converse “if $g_1 * g_2 = g_2 * g_1$, then $a * b = b * a$ for all $a, b \in G$ ” is much more difficult to prove.

Notice that

$$a = a_1 * a_2 * \dots * a_p = \prod_{i=1}^p a_i \quad (3)$$

$$b = b_1 * b_2 * \dots * b_q = \prod_{i=1}^q b_i \quad (4)$$

$$c = a * b = a_1 * a_2 * \dots * a_p * b_1 * b_2 * \dots * b_q \quad (5)$$

where $a_i, b_i \in \{g_1, g_2\}$.

Notice that because $g_1 * g_2 = g_2 * g_1$, we can swap any two adjacent terms of c .

Therefore, we can rewrite c as

$$c = a * b = b_1 * b_2 * \dots * b_q * a_1 * a_2 * \dots * a_p. \quad (6)$$

Therefore, G is generated by g_1 and g_2 then G is abelian if and only if $g_1 * g_2 = g_2 * g_1$. $\square$

- ii. Suppose G is generated by $g_1, g_2, \dots, g_n$. Do you think that G is abelian if and only if $g_i * g_j = g_j * g_i$ for all $i, j \in \{1, 2, \dots, n\}$? Explain your reasoning, but you do not have to provide a proof.

I do think so. We can likely apply some inductive proof to show that adding a new g_k that is mutually commutative with all other g_i does not impact the ‘abelian-ness’ of the overall group G .

- iii. Prove that cyclic groups C_n are abelian.

Proof. By definition,

$$C_n = \langle r \mid r^n = e \rangle.$$

Because the cyclic groups C_n are defined by a generator with one non-identity element, then C_n must be abelian by associativity.

$$r^k * r^l = \underbrace{(r * r * \dots * r)}_k * \underbrace{(r * r * \dots * r)}_l \quad (7)$$

$$= \underbrace{(r * r * \dots * r)}_l * \underbrace{(r * r * \dots * r)}_k \quad (8)$$

$$= r^l * r^k. \quad (9)$$

□

- iv. Prove that the general linear group of 2×2 real-valued matrices

$$GL_2(\mathbb{R}) = \{A \in \mathbb{R}^{n \times n} \mid \det(A) \neq 0\}$$

is not abelian.

Proof. Let

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}, B = \begin{bmatrix} p & q \\ r & s \end{bmatrix}.$$

$$\text{Then, } A \times B = \begin{bmatrix} ap + br & aq + bs \\ cp + dr & cq + ds \end{bmatrix} \text{ and } B \times A = \begin{bmatrix} ap + cq & bp + dq \\ ar + cs & br + ds \end{bmatrix}.$$

In general, these are not the same matrix. □

- v. Prove that the dihedral group D_n is abelian if and only if $n \leq 2$.

Proof. First, by definition, the smallest dihedral group is D_1 , which is just the identity. This trivial group must be abelian.

Then, for $n = 2$,

$$D_2 = \langle r, f \mid r^2 = f^2 = e \rangle.$$

Notice that in this case, $r = f$, therefore, D_2 is abelian.

Then, to prove the “other direction”, we must prove that a dihedral group D_n being abelian implies that $n \leq 2$. We can prove this via contrapositive. Thus, we for a dihedral group D_n where $n > 2$, D_n is not abelian.

Suppose we have D_n where $n > 2$. We choose the element $rf \in D_n$. Now, $rf = fr^{n-1} \neq fr$, so D_n is not abelian.

Therefore, D_n is abelian if and only if $n \leq 2$. □

- (c) The direct product of groups is a way of combining two groups into a new group. The notation can look a bit intimidating, but the basic idea is simple: just treat the groups independently. Here’s the formal definition, but keep in mind that $*$, $\star$, and $\odot$ are just names for binary operations like $+$, $\times$, and $\circ$.

Given two groups $(G_1, *)$ and $(G_2, \star)$, we define the direct product of groups as

$$G_1 \times G_2 = \{(g_1, g_2) \mid g_1 \in G_1 \text{ and } g_2 \in G_2\}$$

with the binary operation, $\odot$, where

$$(g_1, g_2) \odot (g'_1, g'_2) = (g_1 * g'_1, g_2 \star g'_2).$$

- i. Let $(G_1, *) = (D_4, *)$ be the dihedral group of the square, and let $(G_2, \star) = (S_3, \circ)$ be the symmetric group under function composition. Show that in $G_1 \times G_2$,

$$(f, (132)) \odot (r, (23)) = (r^3 f, (13)).$$

$$(f, (132)) \odot (r, (23)) = (fr, (132) \circ (23)) \quad (10)$$

$$= (r^3 f, (2)(31)) = (r^3 f, (13)). \quad (11)$$

- ii. Prove that for any two groups G_1 and G_2 , the direct product of groups is a group.

Proof. Let $G = G_1 \times G_2$. We verify that G follows the group axioms:

Closure: Let $g_1, g'_1 \in G_1$ and $g_2, g'_2 \in G_2$. since $g_1 g'_1 \in G_1$, and $g_2 g'_2 \in G_2$, $(g_1 g'_1, g_2 g'_2) \in G$.

Identity: Let e_1, e_2 be the identity elements of G_1 and G_2 respectively. (e_1, e_2) is the identity of G .

Associativity: Since the operations on the first and second elements of G are both associative, associativity is inherited to the direct product.

Inverse: Similarly, because the left and right elements each independently have an inverse, the tuple containing both will be the equivalent inverse in G .

□

- iii. Prove that if $G_1 \times G_2$ is abelian if and only if G_1 and G_2 are both abelian.

First, if $G = G_1 \times G_2$ is abelian, then for $(a, b), (a', b') \in G$,

$$(a, b) \odot (a', b') = (a', b') \odot (a, b) \quad (12)$$

$$(a * a', b \star b') = (a' * a, b' \star b) \quad (13)$$

Since $a * a' = a' * a$ and $b \star b' = b' \star b$, G_1 and G_2 are abelian.

Now, given G_1 and G_2 are abelian, reverse the above algebra to see that G must also be abelian. Thus, $G_1 \times G_2$ is abelian if and only if G_1 and G_2 are both abelian.

- iv. Let $\mathbb{Z}/k\mathbb{Z}$ denote the group of the integers mod k under addition. Let $G = \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/3\mathbb{Z}$. Compute the order of G . Prove that G has an element of order $|G|$.

The order of a direct product of two groups is the product of the orders. Thus,

$$|G| = |\mathbb{Z}/2\mathbb{Z}| \times |\mathbb{Z}/3\mathbb{Z}| = 2 \times 3 = 6.$$

The element with order 6 is $(1, 1)$.

- v. Let $G = \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$. Compute the order of G . Prove the G has no element of order $|G|$. Similarly, $|G| = 2 \times 2 = 4$. Now, the elements of G are $\{(0, 0), (0, 1), (1, 0), (1, 1)\}$. The orders of all of these elements is 2.

Choice Exercise 7 [10]

- Points attempted: 10
- Self-assessed score: 10/10.
- Time spent: 30 minutes.
- Difficulty: 3/10.
- Questions/Feedback: None.

1. Let G be a group with order n , and denote the order of an element $g \in G$ by $|g|$. Then the order of G is equal to

$$\gcd(|g_1|, |g_2|, \dots, |g_n|) = n.$$

False. The order of group G should be at least the lcm of the orders of the elements.

Answer: False. My reasoning was maybe correct.

2. If $\phi: G \rightarrow H$ is a homomorphism that is 1-to-1, then $\phi: G \rightarrow \text{Im}(\phi)$ is an isomorphism.

True.

Answer: Similar reasoning.

3. If $\phi: G \rightarrow H$ is a homomorphism that is onto, then $\text{Im}(\phi) \triangleleft H$.

False. Consider the case where G is a subgroup of H , but not necessarily a normal subgroup. Now, let $\phi(g) = g$ for $g \in G$.

Answer: True. I overlooked the fact that the homomorphism was onto.

4. If G is a non-abelian group and H is abelian, then the only homomorphism $\phi: G \rightarrow H$ is the trivial homomorphism defined by $\phi(g) = e_H$ for all $g \in G$.

False. Consider the homomorphism from $\phi: D_3 \rightarrow C_2$, where $\phi(r^n) = e$ and $\phi(f) = 1$.

Answer: False. I came to the exact same conclusion, with the same counterexample.

5. If $\varphi: G \rightarrow H$ is an isomorphism, then there always exists an isomorphism $\psi: H \rightarrow G$.

True.

Answer: True, similar reasoning.

6. Let $G \times H$ be a direct product of groups. Then the set

$$N = \{(g, e_H) \mid g \in G\} \subseteq G \times H$$

is a normal subgroup of $G \times H$.

False. This is only a normal subgroup if H is abelian.

Answer: True. I reasoned incorrectly. If I were to correct my reasoning, I would note that $h * e_H = e_H * h = h$.

7. If $G/N \cong H$ then $G \cong N \times H$.

True. Under the isomorphism ϕ , each of the cosets gN can be mapped to a unique $h \in H$. By Lagrange's theory, there must be $|N|$ unique $g \in G$ that maps to each h . Also, gN is closed under $n \in N$. Therefore, we can construct an isomorphism from G to $N \times H$.

Answer: False, my reasoning only showed that the orders of the groups were compatible, not that an isomorphism actually exists.

8. The map $\varphi: \mathbb{Z}/8\mathbb{Z} \rightarrow C_{16}$ given by $\varphi(1) = r \in C_{16}$ is a homomorphism.

False.

Answer: Correct.

9. There is a homomorphism $\psi: D_6 \rightarrow C_6$ such that $\psi(r) = r \in C_6$ and $\psi(f) = e \in C_6$.

False. In D_6 , f can be thought of as “inverting” all subsequent r elements, so this is not a homomorphism.

Answer: Correct, similar reasoning.

10. There is a homomorphism $\phi: C_6 \rightarrow D_6$ such that $\phi(r) = r \in D_6$.

True.

Answer: Correct, but I didn't check the fact that ϕ was structure preserving with the relations. I just wrote out the possibilities.